

Materials Adversarial Forensic Review

Actual project evidence, generated architecture diagrams, and honest scientific claims

Materials Adversarial Learning Project

Generated from repository results

Executive Takeaway

The audit changes the story.

The model is genuinely unstable under token-level perturbations, but the original claim that **length-changing perturbations cause the drift** does not survive forensic testing.

Defensible headline

Real token-level instability exists; the length mechanism is refuted; chemistry-preserving robustness is not yet demonstrated.

Tg samples after cleaning

247

sealed test split

$n = 6$

Transformer parameters

86,337

Problem Statement

Core problem

Can polymer sequence models make stable property predictions under valid-looking PSMILES perturbations, or are observed adversarial effects artifacts of representation, data split, pooling, and attack design?

- ▶ Sequence models can look accurate while learning syntax shortcuts.
- ▶ Token perturbations can change predictions without proving chemical vulnerability.
- ▶ Valid SMILES is not the same as a label-preserving material.
- ▶ The audit asks which claims survive.
- ▶ It separates model instability from causal mechanism.
- ▶ It identifies the experiments still needed before publication.

Research Questions

1. Are polymer sequence regressors vulnerable to valid token-space perturbations?
2. Is the vulnerability specifically caused by length-changing edits?
3. Does adversarial training generalize across attack families?
4. Are current attack labels chemically justified as property-preserving?
5. What exact gaps must be fixed before making a strong robustness claim?

Project Scope and Data

Tg forensic audit focus

- ▶ Raw table: 741 rows, 32 property columns.
- ▶ Tg non-null records: 443.
- ▶ RDKit canonicalization plus conflict dropping.
- ▶ Final Tg dataset: 247 samples.
- ▶ Scaffold split: 204 train / 37 validation / 6 test.

Older bandgap milestone

- ▶ 2,946 train / 631 validation / 632 test.
- ▶ Clean test MAE: 0.4619 eV.
- ▶ Test R^2 : 0.8019.

Deck choice

The Tg forensic audit is treated as the current headline because it supersedes the earlier length-mechanism narrative.

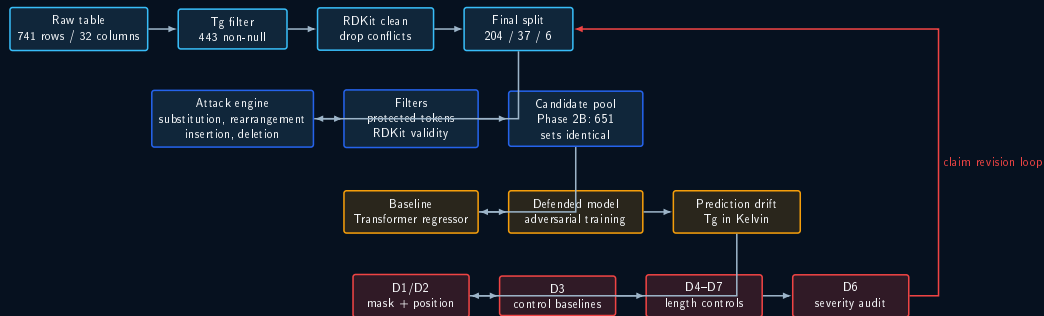
Literature Gap

- ▶ Polymer property prediction often uses SMILES/PSMILES because strings are easy to tokenize.
- ▶ Transformer models can learn token context, but small polymer datasets make shortcut learning likely.
- ▶ Standard evaluations emphasize clean MAE and R^2 more than representation invariance.
- ▶ Adversarial chemistry work often blurs validity, plausibility, and unchanged target labels.

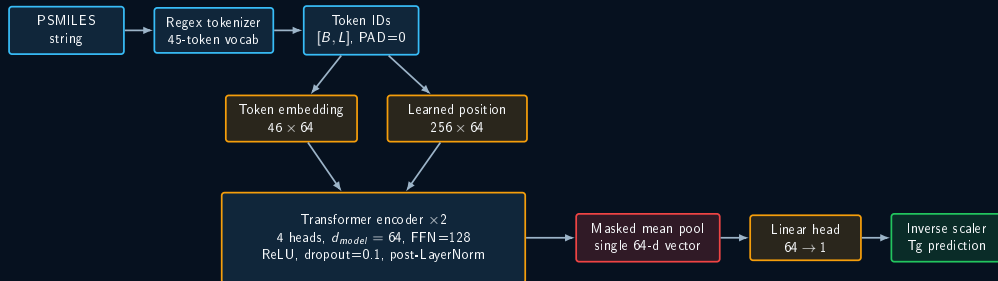
Exact gap this project exposes

The missing control is a representation-level perturbation where the molecule and label are provably unchanged, plus matched-budget evaluation and cross-validated baselines.

Generated System Architecture



Generated Model Architecture



Risk: insertion re-indexes downstream absolute positions

Risk: mean pooling creates $1/L$ dilution

Scale: 86,337 params on 204 Tg train samples

Attack and Evaluation Pipeline

- ▶ Attack families: substitution, rearrangement, insertion, deletion.
- ▶ Candidates pass protected-token constraints, RDKit parsing, plausibility checks, and drift scoring.
- ▶ Phase 2B used identical candidate sets for baseline and defended models.

Pipeline facts

Metric	Value
Phase 2B candidates	651
Unique candidates	651
Candidate sets identical	true
Threshold in Tg audit	52.02 K

Baseline Findings

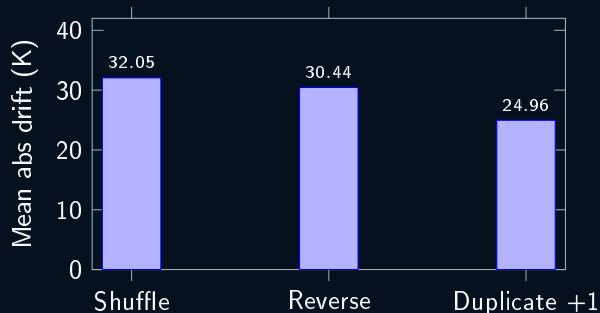
Phase 1 Tg baseline

Split	MAE (K)	R^2
Validation	79.04	-0.150
Test	52.02	-4.246

Interpretation

The model is unstable, but clean predictive strength is fragile in the Tg split. Negative R^2 values make any clean-MAE claim weak, especially with $n = 6$ test samples.

Forensic Result 1: Length Claim Refuted



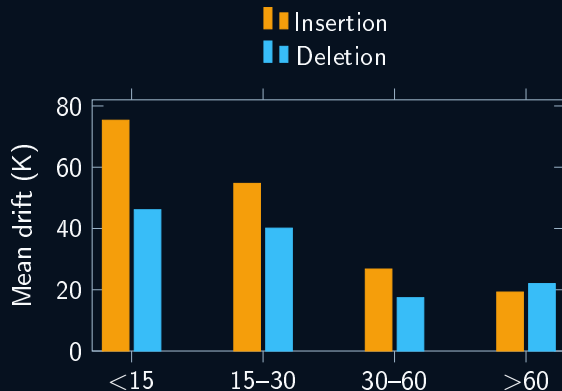
Decisive audit result

Shuffle and reverse preserve length and token multiset, yet produce drift of the same magnitude as length-changing edits across five seeds.

Conclusion

The model is not specifically length-sensitive; it is globally unstable to token-level perturbation.

Forensic Result 2: Mean-Pool Dilution



Observed correlation

- ▶ $\text{corr}(\text{length}, \text{insertion drift}) = -0.502$
- ▶ $\text{corr}(1/\text{length}, \text{insertion drift}) = +0.575$

Meaning

Per-edit drift is confounded by sequence length because the model averages token states.

Forensic Result 3: Attack Families Are Unequal

Family	Valid n	Validity	Mean $ \Delta MW $	Formula kept	Identical	Mean drift
Substitution	88	48.6%	10.79	0.0%	0.0%	12.97 K
Insertion	85	45.9%	15.75	12.9%	12.9%	30.67 K
Deletion	94	50.8%	12.28	2.1%	1.1%	30.73 K
Rearrangement	63	63.0%	0.00	100.0%	1.6%	7.99 K

Interpretation

Rearrangement preserves formula; insertion and deletion change heavy atoms. Family comparisons therefore confound attack mechanism with chemical severity.

Forensic Result 4: Controls Beat the Transformer

Model	Val MAE	Test MAE
Mean predictor	88.48	132.54
Median predictor	87.53	130.67
Length-only linear	76.77	76.39
Bag-of-tokens ridge	79.59	43.24
Transformer	79.04	52.02

Uncomfortable result

A one-feature length-only regression beats the Transformer on validation: 76.77 K vs 79.04 K.

Split confound

$\text{corr}(\text{length}, T_g) = 0.372$ across all 247 samples.

Split Confound

Split	n	Mean Tg (K)	Mean token length
Train	204	330.7	22.6
Validation	37	384.5	43.5
Test	6	463.2	56.7

Consequence

Scaffold splitting sorted longer, higher-Tg polymers into validation and test. The test set is tiny and not representative.

Defended Model: What Worked

Attack family	Baseline mean drift	Defended mean drift	Local change
Substitution	12.97 K	7.02 K	Improved
Insertion	30.67 K	19.29 K	Improved in Phase 2B
Deletion	30.73 K	16.81 K	Improved in Phase 2B
Rearrangement	7.99 K	8.02 K	No improvement

Pooled Phase 2B result

Mean drift dropped by 8.48 K; 63.9% of paired candidates improved under controlled candidate sets.

Defended Model: What Failed

- ▶ Phase 2C used five seeds: 20260815–20260819.
- ▶ Clean model selection remained validation-only; the manifest says the test set was never loaded.
- ▶ The apparent Phase 2A improvement was mostly scaler mismatch; only 23.7% survived controlled decoding.
- ▶ No current attack family is a valid representation-level, label-preserving attack.

Bottom line

Adversarial training may reduce seen-family drift locally, but it does not yet prove chemical invariance or general robust learning.

Research Questions Answered

Question	Answer from the audit
RQ1	Yes. Model instability exists: roughly 20–40 K drift under token perturbation.
RQ2	No. Length is not the mechanism; length-preserving shuffle/reverse are equally disruptive.
RQ3	Partly. Seen-family/local improvements exist, but generalization is not established.
RQ4	No. Current attack families change chemistry or isomerize; none is a clean representation-level attack.
RQ5	Need SMILES randomization, matched budgets, cross-validated baselines, and pooling/position ablations.

Limitations and Exact Gaps

- ▶ Tg test set has $n = 6$; clean-MAE comparisons are anecdotal.
- ▶ Validity-conditioned drift is a survivorship filter with family-specific validity rates.
- ▶ Attack budgets are not matched, and max drift scales with candidate count.
- ▶ No ab initio, synthesis, or laboratory validation confirms true property preservation.
- ▶ No implemented representation-level attack yet; SMILES randomization is the missing control.

Novelty

- ▶ Forensic audit separates real instability from a false length-change mechanism.
- ▶ Architecture diagnosis ties drift to learned absolute positions, masked mean pooling, and small-data training.
- ▶ Chemical-severity audit shows why attack-family comparisons were not controlled experiments.
- ▶ Five-seed controls falsify the Phase 1F causal story.
- ▶ Revised research program prioritizes representation-level invariance before larger defenses.

Next Work

1. SMILES-randomization invariance test: same molecule, provably same label.
2. Positional-encoding ablation across five seeds.
3. k-fold scaffold cross-validation: length-only, bag-of-tokens, Transformer.
4. Budget-matched attacks with median drift, validity rate, and no-op counts.
5. Only then consider graph models, MCMC, or broader defense architectures.

Final Takeaway

Strongest honest claim

We built and audited a materials adversarial framework, found real token-level model instability, and showed that several attractive robustness claims collapse unless representation-level invariance, matched attack severity, and small-data baselines are handled first.

Instability confirmed. Length mechanism refuted. Next experiments are clear.